



JAIPUR NATIONAL UNIVERSITY

End Semester/Year BACK Examination December -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech. (Computer Science)
Semester/Year:	1
Paper Code:	BTEPHCO103T22
Title of Paper:	Engineering Physics
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
2. Section-A consists of seven questions and
3. Section-B consists of six questions.

Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	Show that the apparent length of a rigid body in the direction of its motion with uniform velocity v is reduced by the factor $\sqrt{1 - \frac{v^2}{c^2}}$.	4	3	3
Q.2	Deduce Einstein's mass energy relation and write its significance.	4	5	3
Q.3	Explain Rayleigh's criterion of resolution. Derive an expression for the resolving power of a grating.	4	2	1
Q.4	What are Einstein's coefficients A and B? Obtain the relation between them. Also discuss the essential conditions for laser action.	4	1	2

maxima are nearly $1: \frac{1}{22}: \frac{1}{61}: \frac{1}{121}$.

(b) State and prove Malus law.

Q.4

(a) Draw a neat diagram of Ruby laser and describe its construction and method of working.

10 (5, 5)

1

2

(b) Explain with neat diagram absorption, spontaneous emission and stimulated emission of radiation in LASER.

Q.5

(a) Derive Maxwell's equation.

Explain the physical significance of each equation.

10 (5, 5)

6

5

(b) State and prove the Gauss divergence theorem. Define acceptance angle (AA) and numerical aperture (NA) numerical aperture of optical fiber.

Q.6

(a) A particle of mass m is confined in a field free region between impenetrable walls at $x = 0$ and $x = a$. Show that stationary energy levels and wave function of the particle are given by

10 (7, 3)

4

4

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad \& \psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$$

(b) Find the energy (in eV units) of an electron moving in 1-D in an infinitely high potential box of width $1 \text{ \AA}$.

Q.5	What do you mean by total internal reflection? Discuss propagation of light in multimode step-index and multimode graded index fiber.	4	2	5
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Q.6	State Heisenberg uncertainty principle. Derive the de-Broglie relationship for wavelength (λ) associated with a particle of mass m moving with velocity v .	4	6	4
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Q.7	Derive the Schrodinger's time independent wave equation.	4	4	6
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SECTION - B

Q.1	Derive the Lorentz transformation equations. Under what condition Lorentz transformation equations becomes Galilean transformation equations.	10	3	3
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Q.2	In Newton's ring experiment, prove that in case of reflected light	10		
	(i) diameter of bright rings are proportional to the square roots of odd natural numbers and		2	1
	(ii) the diameters of dark rings are proportional to the square roots of natural numbers.			

Q.3	(a) Derive an expression for the intensity distribution due to Fraunhofer diffraction at a single slit. Obtain the conditions for principal maxima, minima and secondary maxima. Show that the relative intensities of successive	10 (7, 3)	5	1
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